



15-Day Day-wise Plan (Frontend vs Backend Split)

Stack assumed:

Frontend: React + Tailwind + React Router

Backend: Spring Boot + JWT + MySQL/PostgreSQL



Day 1 – Requirements & Architecture



Backend Tasks

- Finalize functional requirements
- Define REST API contracts (endpoints, request/response DTOs)
- Decide database entities
- Draw high-level backend architecture (Controller–Service–Repository)



Frontend Tasks

- Identify UI screens & user journeys (Citizen, Officer, Admin)
- Create basic wireframes (Login, Dashboard, Forms)
- Decide routing structure (public vs protected routes)



Deliverables

- Requirement document
- Architecture diagram
- Screen list + navigation flow



Day 2 – Database & ER Design



Backend Tasks

- Design ER diagram
- Create normalized DB schema
- Define JPA entities & relationships
- Prepare initial SQL migration scripts

Frontend Tasks

- Define UI data models (what fields each form needs)
- Map backend entities to UI forms (e.g., complaint form fields)
- Create mock JSON responses for UI testing

Deliverables

- ER Diagram
- Database schema
- UI field mapping document

Day 3 – Project Setup (Base Infrastructure)

Backend Tasks

- Create Spring Boot project
- Configure:
 - Maven
 - application.properties
 - DB connection
- Setup base packages (controller, service, repo, security)
- Enable CORS

Frontend Tasks

- Create React project
- Install Tailwind, Router, Axios
- Setup base layout (Navbar, Sidebar, Footer)
- Setup routing skeleton

Deliverables

- Backend app running

- Frontend app running
- Base routing working



Day 4 – Authentication & Authorization Module



Backend Tasks

- Implement:
 - Register API
 - Login API
 - JWT token generation
 - Role-based authorization filters
- Secure endpoints with roles



Frontend Tasks

- Build:
 - Login page
 - Register page
- Implement:
 - JWT token storage (localStorage)
 - Auth guard (Protected Routes)
- Handle login errors & loading states



Deliverables

- Secure auth APIs
- Working login/register UI
- Role-based routing



Day 5 – Citizen (Villager) Management



Backend Tasks

- CRUD APIs for:
 - Citizen profile
 - Family details
 - Address
- Validation & exception handling

Frontend Tasks

- Build:
 - Citizen profile form
 - Profile view page
- Integrate APIs with forms
- Show validation errors

Deliverables

- Citizen APIs
- Profile management UI

Day 6 – Village & Panchayat Management

Backend Tasks

- APIs for:
 - Village master data
 - Officer creation
 - Village–Officer mapping
- Admin-only access enforcement

Frontend Tasks

- Admin UI:
 - Add/Edit village
 - Add/Edit officers

- Assign officers to villages

Deliverables

- Village & Officer APIs
- Admin management screens

Day 7 – Schemes & Benefits Module

Backend Tasks

- APIs:
 - Create scheme (Admin)
 - List schemes (Citizen)
 - Apply for scheme
 - Eligibility check
 - Approve/Reject (Officer)
- Status transitions

Frontend Tasks

- Citizen UI:
 - Scheme listing
 - Apply scheme form
- Officer UI:
 - View applications
 - Approve/Reject buttons

Deliverables

- Scheme workflow APIs
- End-to-end scheme application UI

Day 8 – Complaints & Grievance Module

Backend Tasks

- APIs:
 - Submit complaint
 - Assign complaint
 - Update status
 - View complaints by role
- Status lifecycle (OPEN → IN_PROGRESS → RESOLVED)

Frontend Tasks

- Citizen UI:
 - Complaint form
 - Status tracking
- Officer UI:
 - Complaint list
 - Resolution form

Deliverables

- Complaint APIs
- Complaint workflow UI

17 Day 9 – Village Services Module

Backend Tasks

- APIs:
 - Create service request
 - Approve/Reject service
 - Track service status
- Document upload support (optional)

Frontend Tasks

- Citizen UI:
 - Certificate request forms
- Officer UI:
 - Service approval dashboard

Deliverables

- ServiceRequest APIs
- Certificate request UI

17 Day 10 – Announcements & Notifications

Backend Tasks

- APIs:
 - Publish announcement
 - List announcements
- Optional: Email/SMS trigger logic

Frontend Tasks

- Admin UI:
 - Create announcement
- Citizen UI:
 - Announcement feed
 - Highlight emergency alerts

Deliverables

- Announcement APIs
- Announcement UI

17 **Day 11 – Reports & Dashboard**

Backend Tasks

- Aggregation APIs:
 - Total citizens
 - Schemes applied
 - Complaints resolved
- Optimize queries

Frontend Tasks

- Dashboard UI:
 - Summary cards
 - Charts (optional)
- Role-based dashboards

Deliverables

- Dashboard APIs
- Analytics UI

17 **Day 12 – Integration & State Management**

Backend Tasks

- Cross-module integration testing
- Global exception handling
- API documentation (Swagger)

Frontend Tasks

- Global state management (Context/Redux)
- Common components (tables, modals, loaders)
- Error boundaries

Deliverables

- Stable APIs
- Clean, reusable UI components

17 Day 13 – Security, Validation & Hardening

Backend Tasks

- Input validation
- Role enforcement audit
- Rate limiting (basic)
- Logging & audit trail

Frontend Tasks

- Form validation
- Disable unauthorized UI actions
- Improve UX for errors & permissions

Deliverables

- Secured endpoints
- Robust UI validations

17 Day 14 – Testing & Bug Fixing

Backend Tasks

- Unit testing (basic)
- Postman collections
- Fix API bugs

Frontend Tasks

- Manual UI testing

- Cross-browser testing
- Fix UI bugs

Deliverables

- Tested system
- Bug-free demo build

Day 15 – Documentation, PPT & Viva Prep

Backend Tasks

- Prepare:
 - API documentation
 - Architecture explanation
- Final deployment (local/cloud)

Frontend Tasks

- Prepare:
 - UI screenshots
 - Demo flow
- PPT diagrams & demo flow recording

Deliverables

- Final Report
- PPT
- Demo-ready DVMS